

Conclusion: SUPREM3 simulation confirmed by fabrication of prototype experimental devices has shown that a simple junction-isolated backgate electrode, formed by boron implantation through the silicon film and buried oxide, can provide effective dynamic control of V_T for fully depleted SOI CMOS operating from ≈ 1 V power supplies.

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N.G. Tarr, R. Soreefan, W.M. Snelgrove and S. Bazarjani
(Department of Electronics, Carleton University, Ottawa, Canada, K1S 5B6)
T.W. MacElwee (Northern Telecom, PO Box 3511, Station C, Ottawa, Canada, K1Y 4H7)

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GaAs/AlGaAs travelling wave electro-optic modulator with an electrical bandwidth >40GHz

R. Spickermann, S.R. Sakamoto, M.G. Peters and N. Dagli

Indexing terms: Electro-optic modulators, Optical modulation
A GaAs/AlGaAs travelling wave Mach-Zehnder electro-optic modulator with novel slow wave electrodes was fabricated on undoped epitaxial layers. The device has a measured electrical bandwidth of >40GHz at 1.55 μ m.

Introduction: External optical modulators with wide electrical bandwidths are essential components for the low chirp transmission of microwave and millimetre wave signals over optical fibres [1, 2]. In our research we are investigating ways of making modulators with 3dB electrical bandwidths approaching 100GHz. Recently, we reported a travelling wave modulator with a >40GHz electrical bandwidth [3]. However, this device had a V_π of 28V at 1.3 μ m, which was rather high. Here we report a device with a different electrode structure. This modulator also achieves a >40GHz electrical bandwidth, but V_π is reduced by a factor of two.

In the travelling wave configuration, velocity matching between the modulating microwave and modulated optical signals is necessary to maximise bandwidth [1]. In the case of perfect velocity matching, the frequency at which the total electrode loss becomes 6.34dB determines the 3dB electrical bandwidth of the modulator. Therefore, a velocity matched low loss electrode geometry is essential for obtaining wide electrical bandwidths.

In our previous work we designed a velocity matched electrode with 50 Ω characteristic impedance, Z_0 [4]. However, the gap across which the voltage is applied was 9 μ m [3]. This necessitated

large voltages to create the required electric field. Reducing the gap further could reduce the drive voltage, but this also increased microwave loss owing to increased current crowding. Increased microwave loss limited the bandwidth. Therefore, we had to change the electrode structure completely.

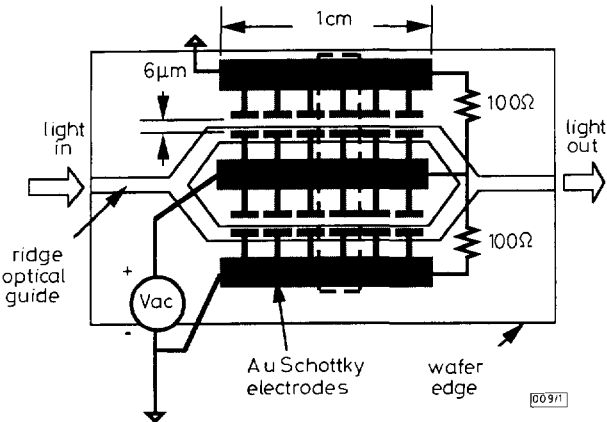


Fig. 1 Schematic top view of modulator

Device description: As seen in Fig. 1, the optical structure is a guided wave Mach-Zehnder interferometer. A specially designed slow wave electrode structure runs parallel with the interferometer. A [100] Al_{0.25}GaAs (1.8 μ m)/GaAs (0.635 μ m)/Al_{0.25}GaAs (0.7 μ m) heterostructure grown using MBE on semi-insulating GaAs provides vertical optical waveguiding. The lateral waveguiding is obtained by etching 0.52 μ m ridges down into the top 0.7 μ m thick Al_{0.25}GaAs layer. The epitaxial structure is unintentionally doped, to keep the optical and microwave losses low.

In III-V compound semiconductors, velocity matching requires a slowing down of the microwave signal. This can be achieved by loading a transmission line periodically using the capacitance of a Schottky-in junction as in [1]. In our case, since the entire epitaxial structure is undoped, the electrode is designed as a slow wave transmission line. As seen in Figs. 1 and 2, it is a modified coplanar line, in which T-rails stem from either side of the centre conductor and from the inner side of both ground planes [5]. These T-rails form tiny capacitors, which periodically load the line and increase its capacitance per unit length. This slows the microwave signal propagating on the electrode. For this structure, axial transmission line currents cannot flow along these T-rails; only displacement current flows in these tiny capacitors. Therefore, current crowding, and the microwave loss, is determined by the distance between the centre conductor and the ground plane of the unloaded line. That gap can be made large to keep microwave loss low [5]. Conversely, the gap between the T-rails determines the field applied to the optical guides for a given voltage. In this design this gap can be reduced significantly with a very small increase in the microwave loss [5]. For a gap as small as 2 μ m measured microwave loss coefficient, α was <4.5dB/cm at 35GHz

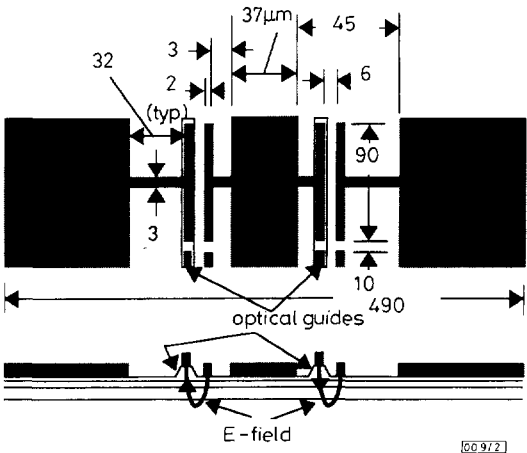


Fig. 2 Top view schematic diagram of modulator section shown by dashed line in Fig. 1, and cross-sectional schematic diagram
All measurements in micrometres

[5]. As a result, the electrode gap has been decoupled from the electrode loss. In the device reported in this Letter, this gap is 6 μm . This design, combined with undoped epitaxial layers, yielded a measured α of 3.0 dB/cm at 40 GHz. A smooth coplanar line with a gap this small would experience excessive current crowding losses and a low electrical bandwidth.

In addition to low loss, the phase velocity v_{ph} is 8.95 cm/ns, which is within 1% of the v_{ph} of an optical wave in GaAs/AlGaAs at 1.3 and 1.55 μm . Furthermore, no dispersion is observed in the v_{ph} up to 40 GHz [5]. The Z_0 of this electrode structure was measured to be $46 \pm 1 \Omega$.

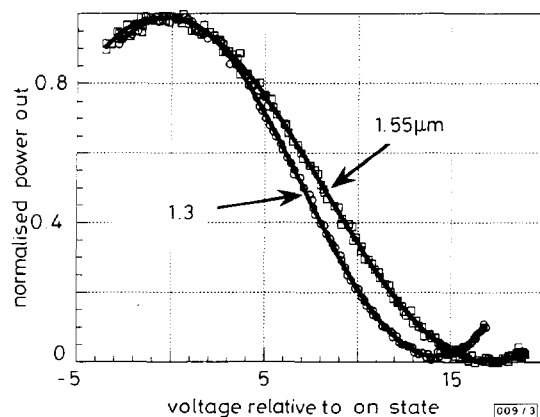


Fig. 3 1 MHz transfer functions of modulator at 1.3 and 1.55 μm

The microwave electrodes are fabricated by evaporating 200 Å/200 Å/1 μm of Ti/Pt/Au. This forms a Schottky contact with the epitaxial layers. A voltage applied between the electrodes biases two back to back Schottky diodes. This makes it possible to apply mainly [100] directed electric fields of opposite polarity on the optical guides [3]. This generates phase shifts of opposite sign on both arms through the linear electrooptic effect, creating a net differential phase shift between the arms of the interferometer. Hence push-pull modulation results.

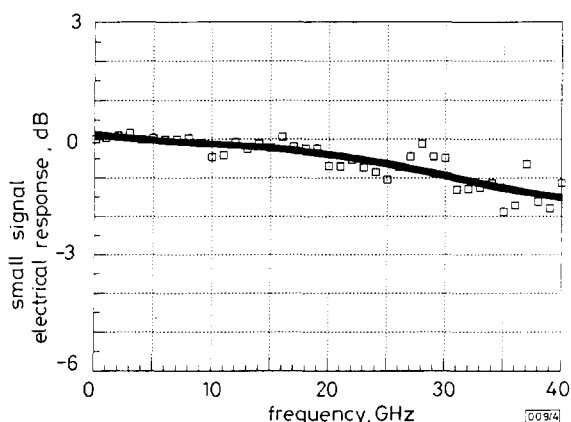


Fig. 4 Modulator small signal response at 1.55 μm

Fitted curve is a fourth order polynomial

Experimental results: The output of a DFB laser was coupled in and out of the waveguide using lensed fibres. The electrical signal was applied using commercially available microwave coplanar probes. Fig. 3 shows the 1 MHz transfer functions of the device at 1.3 and 1.55 μm . The switching voltages are 14 V at 1.3 μm and 16.8 V at 1.55 μm . As predicted, the switching voltage varies inversely with wavelength. The extinction ratio is 20.1 dB at 1.3 μm and 22.6 dB at 1.55 μm . The small signal electrical bandwidth of the device at 1.55 μm , measured using the method outlined in [6], is shown in Fig. 4. The maximum measurement frequency is 40 GHz, owing to equipment limitations. Actual electrical bandwidth should be significantly higher than 40 GHz.

Conclusions: A travelling wave GaAs/AlGaAs electro-optic modulator with an electrical bandwidth > 40 GHz is reported. The device uses unintentionally doped epitaxial layers to keep optical and microwave losses low. A novel slow wave electrode design

makes it possible to obtain velocity matching, a Z_0 of $46 \pm 1 \Omega$, very low microwave loss, a small electrode gap, and hence efficient modulation.

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R. Spickermann, S.R. Sakamoto, M.G. Peters and N. Dagli
(Department of Electrical and Computer Engineering, University of California, Santa Barbara, CA 93106, USA)

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Low order multimode generation in hollow glass waveguides

Y. Matsuura, C.D. Rabii, K. Matsuura and J.A. Harrington

Indexing terms: Optical waveguides, Integrated optics

A mode mixing caused by localised stress in hollow core waveguides is investigated. It is shown that a hollow glass waveguide with a thick glass wall very effectively suppresses low order mode generation resulting from stress, and that a cable made with a thick wall glass waveguide produces a good spatial output mode.

Hollow waveguides are an attractive alternative to solid-core IR fibres for sensor and laser power delivery applications [1]. We have recently developed a new hollow core waveguide by using liquid-phase chemistry methods [2]. Although the losses for the waveguides are quite low [2], a problem often arises when a stress is applied to the guide. Instead of propagating the lowest order LP_{01} mode we find that mode mixing occurs and the spatial purity of the output mode degrades. This is a deleterious effect which impacts upon many power delivery applications of the guides. Other studies have also reported mode mixing in hollow glass [2, 3], metal [4], and plastic waveguides [5]. It was reported that mode mixing can occur in waveguides bent with a relatively large bending radius [4]. However, in small bore guides, with bore sizes $< 500 \mu\text{m}$, we found that the influence caused by stress is much more dominant than the effect of bending [3]. In this Letter we report the first detailed study of mode mixing in hollow waveguides resulting from stress, and show how this effect may be drastically reduced by using guides with thicker glass walls.

For a theoretical evaluation of bent hollow waveguides, a field distribution shift owing to bending which causes radiation losses has been considered [6]. However, in the abrupt, localised bending, it is difficult to evaluate the field distribution owing to the field not being stabilised in the short bending section. Therefore, we consider the mode mixing between the lowest order mode and the